Thursday, January 16, 2025 1:03 PM

Properties of Subspace

Subspace :- Let V be a vector space over F.
A subset, W, $W \neq \varphi$, is called subspace
if (i) $w_1, w_2 \in W \Rightarrow w_1 + w_2 \in W$ ✓
(ii) $w \in W \Rightarrow \alpha w \in W$, $\alpha \in F$. ✓

① Let W_1 and W_2 be two subspace of V.
Is $W_1 \cap W_2$ a subspace of V?

let $w, w' \in W_1 \cap W_2$

$w, w' \in W_1$ | $w, w' \in W_2$
$w + w' \in W_1$ | $w + w' \in W_2$

$w + w' \in W_1 \cap W_2$

$w \in W_1 \cap W_2$

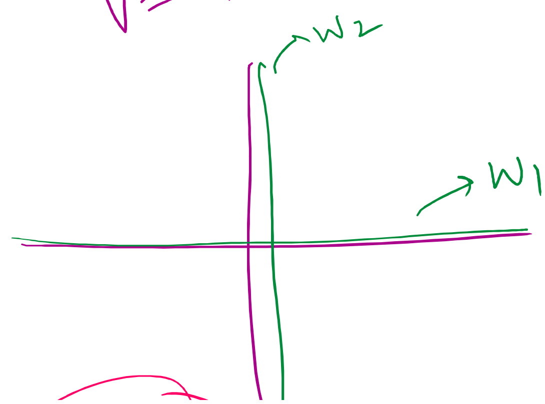
$w \in W_1$ | $w \in W_2$
$\alpha w \in W_1$ | $\alpha w \in W_2$

$\alpha w \in W_1 \cap W_2$

$\therefore$ $W_1 \cap W_2$ is a subspace.

question ② :- Is $W_1 \cup W_2$ a subspace of V??
may not be.

$V = \mathbb{R}^2$



$W_1 = \langle (1,0) \rangle$ = x axis

$W_2 = \langle (0,1) \rangle$ = y axis

$W_1 \cup W_2 = \{ (\alpha, 0), (0, \beta) \mid \alpha, \beta \in \mathbb{R} \}$

$(\alpha, 0) \in W_1 \cup W_2$

$(\alpha, 0) \in W_1 \cup W_2$

$(0, \beta) \in W_1 \cup W_2$

$(\alpha, \beta) = (\alpha, 0) + (0, \beta) \notin W_1 \cup W_2$

$W_1 + W_2$ $\mathbb{R}^2$

Question (3) $V \rightarrow$ vector space over F

$W_1, W_2 \rightarrow$ two subspaces of F

$$W_1 + W_2 = \{ w_1 + w_2 \mid w_1 \in W_1,\ w_2 \in W_2 \} \subseteq V$$

Is $W_1 + W_2$ a subspace of V??

Let $v \in W_1 + W_2$ $\quad v = w_1 + w_2$

$w_1 \in W_1$, $w_2 \in W_2$ $\quad \alpha v = \alpha(w_1 + w_2)$

$\alpha w_1 \in W_1$ $\quad \alpha w_2 \in W_2$ $\quad \alpha v = \underbrace{\alpha w_1}_{W_1} + \underbrace{\alpha w_2}_{W_2} \in W_1 + W_2$.

- $v, v' \in W_1 + W_2$ $\quad v = w_1 + w_2$ $\quad W_1 \Rightarrow w_1 + w_1' \in W_1$

$v' = w_1' + w_2'$ $\quad W_2 \Rightarrow w_2 + w_2' \in W_2$

$$\Rightarrow v + v' = (w_1 + w_2) + (w_1' + w_2') = w_1 + w_2 + w_1' + w_2'$$

$$= \underbrace{(w_1 + w_1')}_{W_1} + \underbrace{(w_2 + w_2')}_{W_2}$$

$W_1 + W_2$

- Let W_1 and W_2 be two subspaces of V. Then linear span of $W_1 \cup W_2$ is $W_1 + W_2$.

$w_1 \in W_1$, $w_2 \in W_2$

$w_1 + w_2 \notin W_1 \cup W_2$ may not

$w_1 + w_2 \in W_1 + W_2$

Direct Sum: Let V be a vector space over F.

Direct Sum :- Let V be vector space over F. Let W_1 and W_2 be two subspaces of V. The vector space V is called direct sum of W_1 and W_2 if

we write $V = W_1 \oplus W_2$

(i) $V = W_1 + W_2$

(ii) $W_1 \cap W_2 = \{0\}$. $\rightarrow$ zero vector.

$\mathbb{R}^2$

$(x,y) = (x,0) + (0,y)$

$x$axis + $y$axis

Example :- ① $V = \mathbb{R}^2$ $\quad W_1 = x$ axis, $W_2 = y$ axis

$x$axis $\cap$ $y$axis $= (0,0)$

② $V = \mathbb{R}^n$ $\quad e_i = (0,0 \_\_ ,0,1,0 \_\_ 0)$ ← i^{th} position

$W_i = \langle e_i \rangle$

$W_1 = \{(\alpha,0,0 \_\_ 0) \mid \alpha \in \mathbb{R}\}$

$W_2 = \{(0,\alpha,0, \_\_ ,0) \mid \alpha \in \mathbb{R}\}$

$\vdots$

$W_n = \{(0,0 \_\_ ,0,\alpha) \mid \alpha \in \mathbb{R}\}$

$\mathbb{R}^n = W_1 + W_2 + \_\_ + W_n$

$W_i \cap W_j = \{0\}$ $\forall$ $i \neq j$, $1 \leq i, j \leq n$

$\mathbb{R}^n = W_1 \oplus W_2 \oplus \_\_ \oplus W_n$

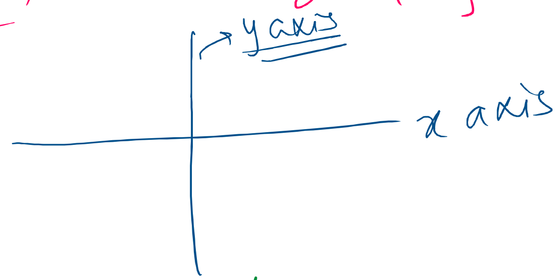
* Let V be a vector space over F. Let W_1 and W_2 be two subspace of V such that $V = W_1 \oplus W_2$, Then every vector $v \in V$ can be written as $w_1 + w_2$,

that

$v \in V$ can be written as $w_1 + w_2$, $w_1 \in W_1$, $w_2 \in W_2$ in a unique way.

$V = W_1 + W_2$, $W_1 \cap W_2 = \{0\}$

$V = \mathbb{R}^2$



$\mathbb{R}^2 = x \text{ axis} \oplus y \text{ axis}$

$(x, y) \in \mathbb{R}^2$

$(x, y) = \underbrace{(x, 0)} + \underbrace{(0, y)}$

Proof :- Let $v \in V$ (By contradiction)

$v = w_1 + w_2$ $\quad w_1, w_1' \in W_1$

$v = w_1' + w_2'$ $\quad w_2, w_2' \in W_2$

$w_1 + w_2 = w_1' + w_2'$

$v' = \underbrace{w_1 - w_1'}_{\in W_1} = \underbrace{w_2' - w_2}_{\in W_2} \Rightarrow v' \in W_1 \cap W_2 = \{0\}$

$\Downarrow$

$v' = 0$

$w_1 - w_1' = 0 \Rightarrow w_1 = w_1'$

$w_2' - w_2 = 0$

$w_2' = w_2$

Let A be a set.

Relation on A is a subset of $A \times A$.

$A = \{a, b, c\}$

$A \times A \supset R = \{(a, a), (a, b), (a, c), (c, b)\}$

$aRa \quad aRb \quad aRc \quad cRb$

$\therefore (b, c) \notin R$

$\Rightarrow$ not related with c.

$\because (b,c) \notin R$
$\Rightarrow b$ is not related with c.

- Equivalence Relation :-

A Relation R on A is Equivalence Relation if it is

① Reflexive
$\Downarrow$
aRa
$\forall a \in A$

② Symmetric
$\Downarrow$
aRb
$\Rightarrow bRa$

③ Transitive
$\Downarrow$
aRb & bRc
$\Rightarrow aRc$

- Equivalence class :- Let R be a relation on set A. & $a \in A$

$$Eq(a) = \{ b \mid bRa \}.$$

- Equivalences classes are either same or disjoint.

Example :- $A = \mathbb{Z} \rightarrow$ Equivalence classes

aRb if $a-b = 5k$, $k \in \mathbb{Z}$

R is Equivalence Relation

transistive $\quad aRb$ & bRc
$\Downarrow \qquad \qquad \Downarrow$
$a-b=5k \qquad b-c=5k'$ $\Big\} \rightarrow a-c = 5(k+k')$
$\Downarrow$
aRc

$Eq(0) = \{ \_\_\_ -10, -5, 0, 5, 10, \_\_\_ \} = 5k = Eq(5)$

$Eq(1) = 5k+1 \qquad Eq(2) = 5k+2 \qquad Eq(3) = 5k+3 \qquad Eq(4) = 5k+4$
$= Eq(6)$

- Let V be a vector space over F. Let W be a subspace of V. Then we

W be a subspace of V. Then

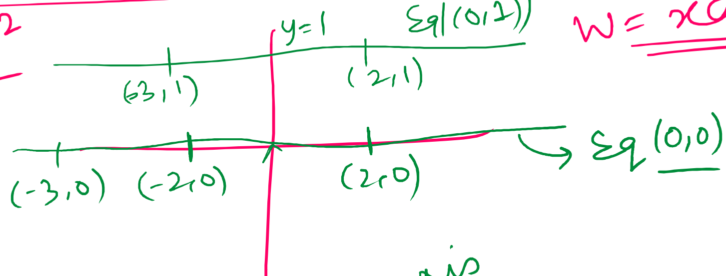
define a relation R on V by

$v_1, v_2 \in V$, $v_1 R v_2$ if $v_1 - v_2 \in W$.

Claim R is an Equivalence Relation on V.

Reflexive	Symmetric	Transitive
vRv $\because v - v = 0 \in W$.	$v_1 R v_2 \Rightarrow v_1 - v_2 \in W$ $\Rightarrow v_2 - v_1 \in W$ $\Rightarrow v_2 R v_1$	$v_1 R v_2 \Rightarrow v_1 - v_2 \in W$ $v_2 R v_3 \Rightarrow v_2 - v_3 \in W$ $v_1 - v_3 \in W \Rightarrow v_1 R v_3$

$V = \mathbb{R}^2$



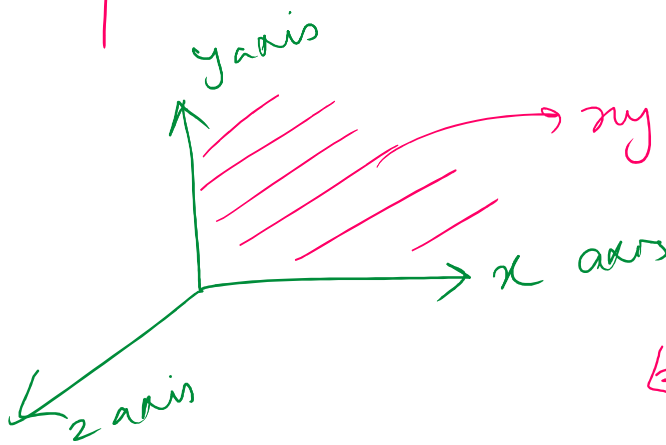
$W = x$ axis

$(x,y)\, R\, (x',y')$

$(x,y) - (x',y') \in x$ axis

$\Leftrightarrow \quad y = y'$

$V = \mathbb{R}^3$



xy plane = W

$(x,y,z)\, R\, (x',y',z')$

$\Leftrightarrow (x,y,z) - (x',y',z') \in xy$ plane

$z = z'$